Comparative Performance Analysis of YOLO Models for Underwater Fish Detection

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Abstract

Accurate detection and counting of fish in underwater video remains challenging in turbid, heterogeneous environments such as coral reefs, where dense schooling, occlusion, and poor visibility degrade detector performance. This study compares two detection paradigms for underwater fish population estimation: a zero-shot open-vocabulary detector (YOLO-World) applied directly to raw frames, and a supervised YOLOv11 model fine-tuned on a manually annotated, domain-specific dataset. From 911 extracted frames, a curated subset was annotated and split 70:20:10 for training, validation, and testing to train YOLOv11, while YOLO-World was evaluated without fine-tuning. Preliminary results show YOLO-World detected 81 fish instances, whereas YOLOv11, even under a trained configuration, detected 742 instances, indicating substantial recall improvement from domain-specific training. The BoT-SORT tracking algorithm was integrated to maintain persistent fish identity across frames despite occlusion and visual similarity within schools. Results highlight the limitations of generic detectors in dense marine scenes and the need for domain-adapted training data, establishing a baseline toward robust fish detection and tracking in complex underwater environments.

1 Keywords

Underwater fish detection, YOLO, Fine-tuning, Multi-object tracking, Coral reef monitoring

2 Introduction

Accurate estimation of fish population, species composition, and individual movement patterns is essential for ecological monitoring, biodiversity conservation, and sustainable fisheries management . Traditional approaches such as manual diver surveys and human-annotated video review are labor-intensive and prone to observer bias, motivating automated, vision-based solutions .

Underwater video presents challenges largely absent in terrestrial detection tasks. Turbidity, variable illumination, and light scattering degrade image quality and object-background contrast . Coral reef environments further contain dense, heterogeneous backgrounds where fish share coloration and texture with coral structures, while large schooling behavior causes frequent inter-object occlusion and visually near-identical instances — conditions under which conventional detectors and trackers, largely developed on human and vehicle datasets, tend to underperform .

Recent advances in real-time object detection, particularly the YOLO (You Only Look Once) family, have demonstrated strong performance due to their favorable speed-accuracy trade-off . Open-vocabulary variants such as YOLO-World extend this by enabling zero-shot detection without task-specific fine-tuning, offering rapid deployment for novel domains such as marine imagery. However, the effectiveness of such zero-shot detectors in visually complex, low-contrast underwater scenes, relative to fine-tuned, domain-adapted detectors, remains underexplored. Similarly, the reliability of multi-object tracking algorithms such as BoT-SORT under dense schooling and frequent occlusion has not been systematically evaluated for underwater fish populations.

We investigate a comparative pipeline for automated fish population estimation, species identification, and persistent tracking in turbid, heterogeneous marine environments.This study evaluates a zero-shot open-vocabulary detector (YOLO-World) against a supervised, fine-tuned YOLOv11 model trained on a manually annotated, domain-specific dataset extracted from coral reef footage.To maintain individual fish identity across frames despite occlusion, this framework integrates the BoT-SORT tracking algorithm with the detection

pipeline, enabling a controlled comparison between zero-shot deployment and domain-adapted supervised training.

Our study aims to address the following research questions (RQ):

- **RQ1:** How does a zero-shot open-vocabulary detector (YOLO-World) compare to a fine-tuned, domain-specific detector (YOLOv11) in detection recall and reliability in turbid, coral-dense underwater environments?
- **RQ2:** To what extent does domain-specific manual annotation and supervised fine-tuning improve fish detection performance over direct zero-shot inference?
- **RQ3:** How effectively does the BoT-SORT tracking algorithm preserve individual fish identity under dense schooling, occlusion, and limited inter-instance visual distinctiveness?

3 Dataset Statistics

The dataset used in this study was provided by NITTTR Chennai, consisting of a continuous 30-second underwater video captured at 30 frames per second (fps), depicting fish movement in a turbid, coral-dense marine environment. Frames were extracted from the video using OpenCV, yielding a total of 911 frames, which were manually annotated using Roboflow with bounding box labels for fish instances.

The annotated frames were partitioned into training, validation, and testing splits in a 70:20:10 ratio, resulting in 638 frames for training, 182 frames for validation, and 91 frames for testing. Each frame frequently contains multiple co-occurring fish instances due to the dense schooling behavior exhibited in the source video, yielding a total of 742 annotated fish instances across the dataset.

4 Computer Vision Model

Computer vision is a subfield of artificial intelligence concerned with enabling machines to interpret and extract meaningful information from visual data such as images and video. Among its core tasks, object detection involves both localizing objects within an image via bounding boxes and classifying them into predefined categories. Deep learning-based detectors, particularly convolutional neural network (CNN) architectures, have substantially advanced the accuracy and efficiency

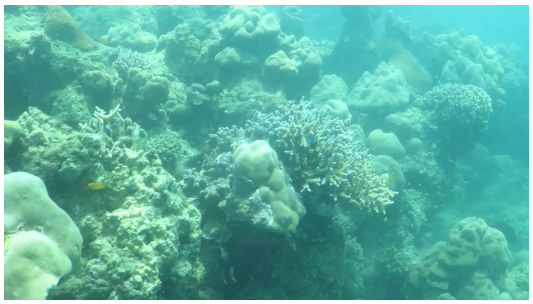


Figure 1: Representative frame extracted from the input video, depicting a turbid, coral-dense underwater environment with multiple small fish partially camouflaged against the reef substrate

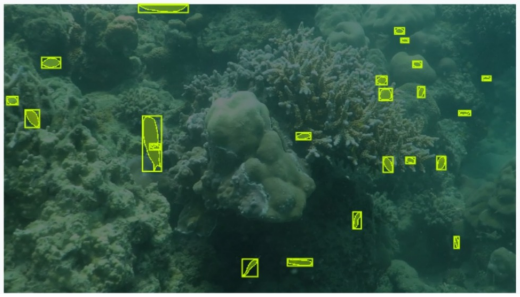


Figure 2: Manual annotation of fish instances in a sample frame using Roboflow, showing bounding boxes for multiple co-occurring, densely distributed fish partially camouflaged against the coral reef substrate.

of object detection across diverse domains, including autonomous navigation, surveillance, and ecological monitoring.

4.1 YOLOv11

YOLO (You Only Look Once) is a family of single-stage object detection architectures that formulate detection as a single regression problem, enabling real-time inference with competitive accuracy. YOLOv11, the latest iteration in this lineage, introduces architectural refinements to its backbone and detection head, improving feature extraction and localization accuracy while maintaining computational efficiency. Owing to its favorable speed-accuracy trade-off, YOLOv11 is well-suited for supervised, domain-adapted detection tasks such as underwater fish detection, where fine-tuning on annotated, task-specific data is required to achieve reliable performance.

4.2 YOLO-World

YOLO-World extends the YOLO architecture to support open-vocabulary object detection, enabling zero-shot detection of arbitrary object categories specified via natural language prompts, without

requiring task-specific fine-tuning. By aligning visual and textual representations during pretraining, YOLO-World can generalize to novel domains and object categories not explicitly seen during training. This capability offers a rapid deployment pathway for data-scarce applications; however, its detection reliability in visually complex, low-contrast environments, such as turbid underwater scenes, remains comparatively underexplored relative to fine-tuned, domain-adapted detectors.

4.3 Model Description

Both detectors were implemented using the `ultralytics` Python package. YOLO-World was employed in its pretrained form, using the class prompt *"fish"* to localize instances directly on raw frames without task-specific fine-tuning. YOLOv11, in contrast, was fine-tuned on the annotated training split (Section 3) to adapt the model to the domain-specific characteristics of the underwater coral-reef environment, including turbidity, dense schooling, and partial occlusion. To preserve individual fish identity across frames despite frequent occlusion and visual similarity within schools, the BoT-SORT multi-object tracking algorithm was integrated with the YOLOv11 detection pipeline for persistent identity association.

4.4 Adaptation Strategies

Each detection paradigm was evaluated under a distinct adaptation strategy, reflecting the trade-off between rapid zero-shot deployment and domain-specific supervised training.

4.4.1 YOLO-World Zero-Shot

YOLO-World, implemented via the `ultralytics` Python package, was evaluated in a zero-shot setting without any task-specific fine-tuning. Raw video frames were passed directly to the pretrained model, with the open-vocabulary class prompt *"fish"* used to localize instances. This approach leverages the model's pretrained visual-semantic grounding to detect fish without requiring annotated training data, serving as a rapid-deployment baseline for underwater detection.

4.4.2 YOLOv11 Training

For supervised adaptation, the YOLOv11n segmentation variant (`yolo11n-seg`) was fine-tuned on the manually annotated training split described in Section 3. The model was trained for 50 epochs with a batch size of 16, an input image size of 640×640, using the Ultralytics default optimizer with early stopping applied via a patience of 10 epochs. Training was performed on an NVIDIA T4 GPU. To preserve individual fish identity across frames despite frequent occlusion and visual similarity within densely schooling fish, the BoT-SORT multi-object tracking algorithm was integrated with the trained YOLOv11 detection pipeline for persistent identity association.

4.5 Implementation

Frames were extracted from the source video using OpenCV and annotated using Roboflow, with bounding box labels exported in YOLO format. All experiments, including YOLO-World zero-shot inference and YOLOv11 training, were conducted in the Google Colab free-tier environment, utilizing an NVIDIA T4 GPU (16GB VRAM).

5 Results

The two detection paradigms – zero-shot YOLO-World and fine-tuned YOLOv11 – are empirically compared based on total fish detections on the source video. Their effectiveness can be analyzed by answering the research questions (ARQ):

- **ARQ1:** From Table 1, it is evident that the fine-tuned YOLOv11 model substantially outperforms the zero-shot YOLO-World model in detection recall, detecting 742 fish instances compared to 81 by YOLO-World. This indicates that generic open-vocabulary detectors, while requiring no annotation effort, struggle to reliably localize fish in turbid, coral-dense underwater scenes.

- **ARQ2:** The substantial increase in detected instances (an approximate 9-fold improvement) demonstrates that domain-specific manual annotation and supervised fine-tuning considerably improve detection performance over direct zero-shot inference, confirming the necessity of task-adapted training data for underwater fish detection.

- **ARQ3:** Integration of the BoT-SORT tracking algorithm with the YOLOv11 pipeline enabled persistent identity association across frames; however, label swapping and identity switches were observed under conditions of dense schooling and prolonged occlusion, indicating limitations in BoT-SORT's

re-identification memory for visually near-identical fish instances.

Table 1: Comparison of fish detection counts between zero-shot YOLO-World and fine-tuned YOLOv11.

Model	No. of Fishes Detected
YOLO-World (Zero-shot)	81
YOLOv11 (Fine-tuned)	742

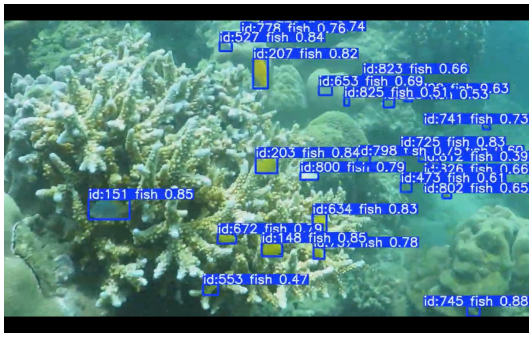


Figure 3: Detection and tracking output on a densely schooling frame, showing YOLOv11 detections with BoT-SORT-assigned identity labels and confidence scores. The wide range and rapid growth of assigned IDs (e.g., id:778, id:825) illustrate frequent identity switching caused by dense fish schooling and visual similarity between instances.

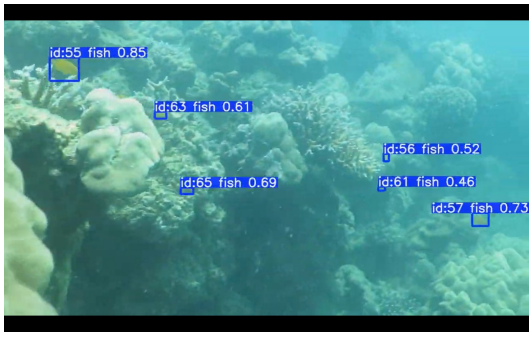


Figure 4: Detection and tracking output on a sparsely populated frame, showing consistent, low-valued identity labels (e.g., id:55–65) assigned by BoT-SORT, reflecting more stable tracking under reduced fish density and occlusion.

6 Conclusion

This study presented a comparative evaluation of zero-shot YOLO-World and fine-tuned YOLOv11 for underwater fish detection using a manually annotated coral reef dataset. The fine-tuned YOLOv11 model demonstrated a considerable improvement in detection capability over the zero-shot approach, highlighting the importance of domain-specific training for complex underwater environments. BoT-SORT was employed for fish tracking; however, identity switching was observed in densely populated scenes due to occlusions and the similar appearance of fish. The findings establish a comparative baseline for future improvements in underwater fish detection and tracking.

7 Limitations

The study was conducted using a continuous 30-second underwater coral reef video, limiting the diversity of environmental conditions and fish species considered. The dataset contained dense schools of fish, resulting in frequent occlusions and label-switching during tracking. In addition, the experiments were performed using the lightweight YOLOv11n model under the computational constraints of the Google Colab free-tier environment. Future work will focus on larger and more diverse datasets, architectural enhancements, and improved tracking techniques to achieve more reliable underwater fish detection and persistent identity preservation.

8 Ethical Considerations

This work uses underwater video data provided by the project supervisor, captured through passive, non-invasive video recording, for academic and research purposes. The dataset contains no human subjects, and no physical intervention with marine organisms or their habitat was involved. As the proposed detection and tracking pipeline is intended to support ecological monitoring, automated outputs such as fish counts and tracking trajectories may contain false positives, false negatives, or identity-switching errors, particularly under dense schooling and turbid conditions; such outputs should therefore be validated by domain experts before use in ecological assessment or conservation decision-making.

Limited AI-assisted writing tools were used to improve the grammatical clarity and presentation of the manuscript. All experimental design, implementation, analysis, and interpretation of results were conducted and verified by the authors. The authors carefully reviewed and validated all AI-assisted content to ensure accuracy, originality, and compliance with academic integrity standards.

9 References

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A Example Appendix

Table 2: YOLOv11 Training Configuration

Parameter	**Value**
Model	YOLOv11n-Seg
Input Image Size	640×640
Epochs	50
Batch Size	16
Device	NVIDIA T4 GPU (CUDA)
Early Stopping (Patience)	10
Project Directory	fish_project
Experiment Name	fish_model_v1

Table 3: Fish Tracking Configuration

Parameter	**Value**
Tracker	BoT-SORT
Detection Confidence	0.25
IoU Threshold	0.50
Identity Persistence	Enabled
Annotated Video Saving	Enabled
Project Directory	fish_count
Experiment Name	track_results